

LensGuide

Visual Search & AR Travel Companion

Snap a landmark, sign, or menu — get an instant, location-aware answer in your own language.

Snap → Identify

Translate 3 Languages

Nearby Landmarks

Team Reboot Rebels · Cambridge Institute of Technology

Every traveller hits the same wall

A tourist with a camera, but no context

Identify landmarks

That monument is ... what, exactly?

Read signs & menus

Cannot read the local script at all.

Find what's near me

No idea where best things are from where I stand.

Language barrier

English ↔ Kannada ↔ Telugu — nowhere, offline.

Trustworthy info

Generic travel sites, not grounded facts.

What the 24-hour data gives us

60 classes

labelled landmarks × 10 images each (train/eval split)

200

menu/sign images with ground-truth text + reference translations

900

grounded POI facts with honest confidence bands

3 languages

BCP-47 only: en-IN · kn-IN · te-IN (Rule R6)

₹0.00

paid-API budget → local-first AI (LM Studio)

One camera flow, eight MVP capabilities

1 Snap → Identify

Camera/upload to landmark, dish, sign or food with location-aware matching

2 Visual Search

Local vision model recognises real-world landmarks — not just the catalogue

3 Translate · 3 languages

Signs & menus → English (en-IN), Kannada (kn-IN), Telugu (te-IN)

4 Nearby Landmarks

Top 10 places ranked by real device GPS (haversine), tap to open

5 Grounded Info Cards

POI facts (poi_facts_kb) + knowledge chunks (place_kb) — honest confidence

6 Snap-to-Book

Hours, cost (currency-exponent-aware), accessibility, carbon — one tap

7 Offline Demo Mode

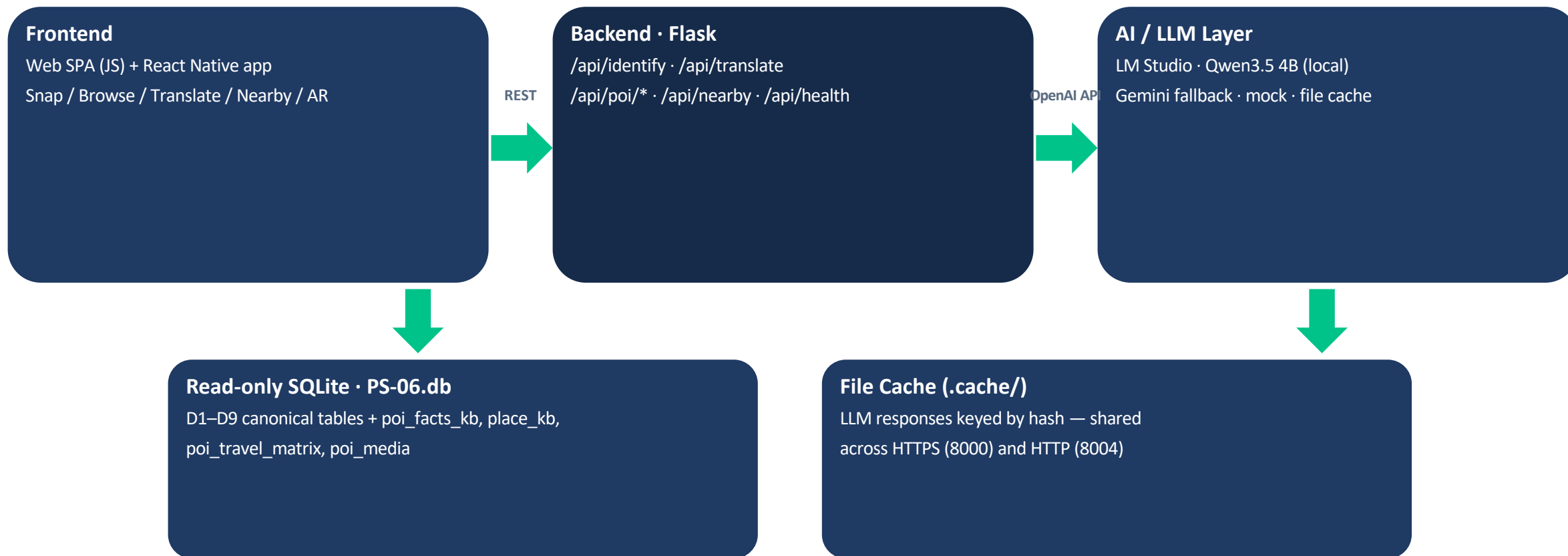
Full POI picker works with zero AI keys

8 Local-First AI

Qwen3.5 4B on LM Studio — no paid APIs, fully on-device

Beyond this MVP ► AR live overlay · 2.5D routes · 3D landmark models (xr_scenes)

Clean three-tier design, AI at the edge



GPS → identify: POI within 2 km = catalogue match · beyond 2 km = visual LLM ID + nearby from real device location

Local-first AI: deterministic, grounded, honest

Provider chain — openai (local) → gemini → mock



Tuning for a 4B local model

- `enable_thinking=false` → fast, token-efficient
- Images downscaled to 768 px (context-safe)
- 512 max tokens · JSON-only prompts
- In-memory + file cache keyed by message hash

1 Identify (vision)

JSON {kind, name, confidence_score, text}

2 OCR text

Extracted from sign / menu / lyrics

3 Translate

en-IN direct · kn-IN / te-IN via English pivot

4 Validate script

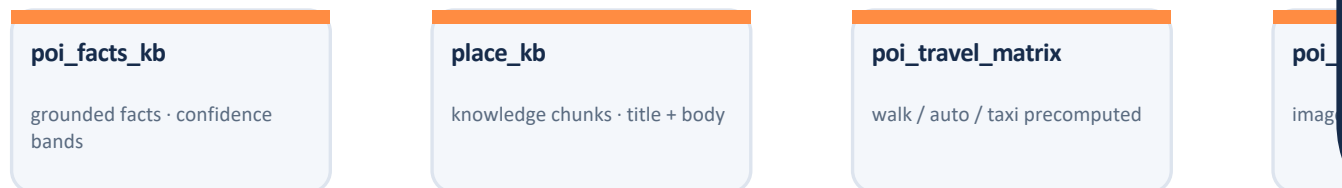
Kannada → ಕನ್ನಡ · Telugu → తెలుగు blocks;
retry if wrong

English pivot: local 4B can't translate kn↔te directly — translate to English first, then to target script. Script validation fails → retry, keeps output native.

Built on the official PS-06 data model



PS-06 extensions added on top



Boundary rules enforced in code

- R1 DB opened read-only (mode=ro) — never mutated
- R2 poi_id treated as opaque strings
- R3 currency minor_unit_exponent → correct display (¥, ₹, KD)
- R6 BCP-47 only — exactly en-IN · kn-IN · te-IN accepted
- R7 WGS-84 coordinates + haversine distance

13 tables · 12,289 rows · PS-06.db (read-only)

Schema: data-model/schema.sql · Seeds: data-model/seed/ (13 CSVs) · Mapping: data-model/DATA_MODEL.md

Five taps from photo to answer

1 Open & Allow GPS

Snap tab · camera/upload · location shared

2 Snap → Identify

Landmark / sign / menu / dish recognised by vision LLM

3 Get the Place

Nearest catalogue POI (≤ 2 km) or live visual ID (+ name)

4 Explore Nearby

Top-10 landmarks by real GPS distance — tap to open

5 Read Info Card

Grounded facts, knowledge, hours, cost, accessibility

6 Translate

Same photo → Kannada / Telugu / English native script

Verified end-to-end: Kannada lyric screenshot → translate to ಕನ್ನಡ / తెలుగు / English with correct native script, plus identify with Bangalore GPS (10 nearby landmarks).

Proof, tests, and what's in the repo

Test suite (11 tests)

- Provider fallback on rate-limit & quota-exhaustion (429 contract)
- Cache prevents repeat LLM calls; transport errors → structured LLMError
- Only en-IN / kn-IN / te-IN accepted (Rule R6)
- Dataset-reference fallback when AI quota is exhausted

Data conformance & runtime health

- ✓ conformance tool passes D1–D9 + extensions on PS-06.db
- ✓ /api/health → ok:true · local_ai:true · 900 POIs indexed
- ✓ train/eval split respected; DB never mutated (R1)
- ✓ live: HTTPS :8000 (web) · HTTP :8004 (app) · LM Studio :1234

Repository (required hackathon layout)

frontend/	web SPA (src/) + React Native app (expo-app/)
backend/	Flask — app.py · llm.py · db.py + requirements.txt
data-model/	schema.sql · 13 seed CSVs · DATA_MODEL.md
ai/	prompts/ (identify.md, translate.md) · pipeline/llm.py
docs/	ARCHITECTURE.md (component diagram, data flows)
tests/	test_llm_fallback.py — 11 tests
scripts/	start_local_ai.sh · serve.sh
README + .env.example	8-section contract + all env vars documented

github.com/challalokesh08/LensGuide · main · 8/8 slides verified